Deval Mehta, PhD

Researcher in Artificial Intelligence for Healthcare and Digital Health

Senior Lecturer in Digital Health at the School of Computing, RMIT University

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Education

PhD, School of Electrical and Electronics Engineering — 2014-2019
Nanyang Technological University, Singapore

BTech, Department of Electrical and Electronics Engineering — 2010-2014
National Institute of Technology, Trichy, India

Work and research Experience

Senior Lecturer	School of Computing, RMIT University	(joining) June 2026
Honorary Research Fellow	Alfred Health	Mar 2022 - present
Research Fellow	Faculty of IT, Monash University	May 2021 - May 2026
Founding Member	AIM for Health Lab, Monash University	July 2023 - May 2026
Research Scientist	IBM Research Labs Australia	Feb 2020 - Apr 2021
Senior R&D Engineer	Panasonic R&D Center Singapore	Apr 2019 - Nov 2019
R&D Engineer	Panasonic R&D Center Singapore	Jul 2018 - Mar 2019

Selected Publications

19 Q1 and CORE A*/A author publications across technical AI conferences and journals, and clinical translation journals. For full list, please refer to Google Scholar Profile. Some of the highlighted works are described below:

- Yiwen, J., D. Mehta, Yan, S., Ge, Z., "WISE: Weak-Supervision-Guided Step-by-Step Explanations for Multimodal LLMs in Image Classification", EMNLP (Main), August 2025. (Top conference in Natural Language Processing).
- Yiwen, J., D. Mehta, W. Fei, Ge, Z. ,"Enhancing Interpretable Image Classification Through LLM Agents and Conditional Concept Bottleneck Models", ACL (Main), June 2025. (Top conference in Natural Language Processing).
- Pham, K., D. Mehta, Verspoor, K., Kwan, P., Ge, Z. ,"Knowledge Tree Driven Contextualized Instruction Tuning of Foundation Models for Epilepsy Drug Recommendation", MICCAI, June 2025. (Top conference in Medical AI). (AI for Epilepsy)
- Yang, H., George, Y., D. Mehta, Lin, L., Yang, D., Ge, Z. , "Multi-modal CT Perfusion-based Deep Learning for Predicting Stroke Lesion Outcomes in Complete and No Recanalization Scenarios", American Journal of Neuroradiology, June 2025. (Top Q1 journal in Neurology). (AI for Stroke).
- R. Kumar, R. Singhal, P. Kulkarni, D. Mehta, K. Jadhav ,"M3CoL: Harnessing Shared Relations via Multimodal Mixup Contrastive Learning for Multimodal Classification", Transactions on Machine Learning Research, May 2025. (Top Q1 journal in Machine Learning).
- D. Mehta, Yiwen, J., Catherine, T., Mingguang, H., Kshitij, J., Ge, Z. ,"Interpretable Few-Shot Retinal Disease Diagnosis with Concept-Guided Prompting of Vision-Language Models", Informa-

tion Processing in Medical Imaging, Feb 2025. (Top conference in Medical AI since 1969). (AI for Eye Disease)

- D. Mehta, Primerio, C., Gal, Y., Mar, V., Soyer, P., Ge, Z. ,“Large Multi-task Vision AI Model for Dermatology: Overcoming Critical Challenges in Skin Lesion Diagnosis”, Journal of the European Academy of Dermatology and Venereology, Jan 2025. (Top Q1 clinical journal in skin disease, acceptance rate: 10%). (AI for Skin)
- Zhang, X., D. Mehta, Zhu, C., Darby, D., Ge, Z., Helmut, B. ,“Adaptive Transformer Modelling of Density Function for Nonparametric Survival Analysis”, Journal of Machine Learning, Jan. 2025. (Top Q1 journal in Machine Learning & AI) (AI for Multiple Sclerosis)
- D. Mehta, Sivathamboo, S., Simpson, H., Kwan, P., Ge, Z. ,“Privacy-Preserving Early Detection of Epileptic Seizures in Videos”, MICCAI, 2023. (Top conference in Medical Imaging)
Top 15% paper early accept out of 2,500. (AI for Epilepsy)
- Asif, U., D. Mehta, Von Cavallar, S., Tang, J., Harrer, S. ,“DeepActsNet: A deep ensemble framework combining features from face, hands, and body for action recognition”, Pattern Recognition, 139, 109484. 2023. (Top Q1 journal in AI).
- D. Mehta, Gal, Y., Bowling, A., Bonnington, P., Ge, Z.,“Out-of-distribution detection for long-tailed and fine-grained skin lesion images”, MICCAI, 2022.
Top 15% paper early accept out of 2,000. (AI for Skin)
- D. Mehta, Asif, U., Hao, T., Bilal, E., Von Cavallar, S., Harrer, S., Rogers, J., “Towards automated and marker-less Parkinson disease assessment: predicting UPDRS scores using sit-stand videos”, CVPR (pp. 3841-3849) 2021. (Top conference in AI). (AI for Parkinson’s)

Research Grants

» Awarded Grants: Sum total - AUD $1,755,036

- D. Mehta, “Advancing Cross-national Digital Health Collaboration: Improving Skin Care Through AI-powered System for Rural Communities in Australia and India”, AUD $13,500, **Australian Academy of Science Australia–India Strategic Research Fund** 2026.
- Kwan, P., D. Mehta, Ge. Z., “MRI Foundation Model for Neurology Imaging and Clinical Translation”, AUD $96,700 [32K A100 GPU hours], **NVIDIA Academic Grant** 2025-26.
- Ge, Z., D. Mehta, Kwan, P., Meng, L., “Neuroimaging Foundation Model Infrastructure for Research and Clinical Translation”, AUD $943,325, **National Imaging Facility (NIF) Infrastructure Grant** 2025-27.
- D. Mehta, Foster, E., Kwan, P., “A Digital Risk Stratification Tool to Reduce Repeat Seizure Presentations to Emergency Departments”, AUD $50,000, **AI in Health Seed Grant Scheme** 2025-26.
- Liu, J., Haryanto, A., D. Mehta, Satriadi, K., “HandovAR: A Framework of AI-facilitated Handover System via Augmented Reality for ICU Nurses”, AUD $30,000, **Faculty of IT Seed Grant** 2025-26.
- Ge, Z., George, Y., D. Mehta, “Collaboration with Optiscan Imaging”, AUD $564,540, **Digital Health CRC-P** 2024-27.
- George, Y., D. Mehta, Kwan, P., “Multimodal Federated Learning for Epilepsy Care and Management”, AUD $34,380, **Faculty of IT Seed Grant** 2024-25.
- Pei, X., Shin, W., D. Mehta, “Comprehending the challenges and opportunities faced by Australian elderly communities in medical AI adoption”, AUD $19,291, **University of Melbourne Arts Collaborative Research Seed Fund** 2022-23.

Graduate Research Supervision

- Project Lead supervisor of 3 successfully graduated PhD students at Monash and 4 ongoing PhD students at Faculty of IT, Monash University.
- Associate supervisor of 2 ongoing medical PhD students at Faculty of Medical Nursing and Health Sciences (MNHS), Monash University.
- Associate supervisor of 1 ongoing Master Student at IIT-Bombay.
- Project Lead supervisor of successful completion of 5 Final Year Project and 2 Master by Minor Thesis students.

Patents

Commercial utility of my research is demonstrated through below intellectual property.

- Deval Mehta, Talha Ilyas, Zongyuan Ge, "Method and System for De-identified Motion Tracking", **Australia Provisional Patent** 2025905326, Nov 2025.
- Deval Mehta, Yiwen Jiang, Zongyuan Ge, Minguang He, Jason Sun, "Multimodal Data Collection And Diagnosis Of Rare Ocular Diseases", **US Patent App.** WO2024US59516, Jan 2024.
- Deval Mehta, Toan Ngyuen, Yaniv Gal, Zongyuan Ge, Paul Bonnington,"Out of Distribution Detection for Long-tailed and Fine-grained Datasets", **New Zealand Patent App.** 786708, Mar 2022.
- Umar Asif, Deval Mehta, Stefan von Cavallar, Stefan Harrer, Jianbin Tang, Tian Hao, Jeffrey L. Rogers, Erhan Bilal,"Privacy-preserving motion analysis", **US Patent App.** 17/249,132, Jan 2021.
- Tian Hao, Jeffrey L. Rogers, Umar Asif, Erhan Bilal, Deval Mehta, Stefan Harrer, Jianbin Tang, Stefan von Cavallar,"Assessing parkinson's disease symptoms", **US Patent App.** 17/249,412, Aug 2020.
- Deval Mehta, Shoushun Chen, and Kay Soon Low,"A High Accuracy Star Tracker Using Running Sequential Angular Match Technique", **Singapore Copyright**, 10201603223W, April 2016.

Academic Fellowships & Achievements

- Early Career Researcher Award, Australia-India Strategic Research Fund, Australian Academy of Science, 2026
- One of the top 3 Finalists for the Victoria Melanoma Young Researcher Award (Morgan Mansell Prize), 2023
- Selected as one of the top 200 early career scientist globally to attend the 10th Heidelberg Laureate Forum (HLF), 2023
- Awarded the HLF travel grant from Australian Academy of Science for attending the 10th HLF (Only 1 in Australia selected every year), 2023
- Awarded TOSHIBA Global Intern at Toshiba Devices & Storage Corporation Tokyo, Japan, 2018
- Nanyang Technological University Research Scholarship for PhD, 2014-2018
- DAAD WISE Scholarship Technical University of Munich, Germany, 2013